

**GUJARAT TECHNOLOGICAL UNIVERSITY****BE- SEMESTER-VI EXAMINATION – WINTER 2025****Subject Code:3161005****Date:17-11-2025****Subject Name:Fiber Optic Communication****Time:02:30 PM TO 05:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

|            |                                                                                                                                                                                                                                                                                                                  | <b>MARKS</b> |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>Q.1</b> | <b>(a)</b> Define: (1) Refractive index of a material.<br>(2) Critical angle with figure.<br>(3) Total internal reflection.                                                                                                                                                                                      | <b>03</b>    |
|            | <b>(b)</b> Describe single mode fibers and multimode fibers with necessary figures and write the equation for number of modes supported by multi mode fiber.                                                                                                                                                     | <b>04</b>    |
|            | <b>(c)</b> Explain the advantages of fiber optic communication system in detail.                                                                                                                                                                                                                                 | <b>07</b>    |
| <b>Q.2</b> | <b>(a)</b> Define V number or Normalized frequency with equation.                                                                                                                                                                                                                                                | <b>03</b>    |
|            | <b>(b)</b> A graded index fiber has a core with a parabolic refractive index profile which has a diameter of 45 $\mu\text{m}$ . The fiber has numerical aperture of 0.25. Estimate the total number of guided modes propagating in the fiber when it is operating at a wavelength of 1.5 $\mu\text{m}$ .         | <b>04</b>    |
|            | <b>(c)</b> Describe propagation of a light signal through a single mode step index fiber and multi mode step index fiber with necessary figures.                                                                                                                                                                 | <b>07</b>    |
|            | <b>OR</b>                                                                                                                                                                                                                                                                                                        |              |
|            | <b>(c)</b> What is dispersion in fiber optic communication?<br>Describe: (1) Material dispersion (2) Wave guide dispersion in detail.                                                                                                                                                                            | <b>07</b>    |
| <b>Q.3</b> | <b>(a)</b> Define skew rays with the help of a figure.                                                                                                                                                                                                                                                           | <b>03</b>    |
|            | <b>(b)</b> An optical fiber with refractive index of core 1.50 and the refractive index of cladding is 1.45, calculate:<br>(1) Critical angle (2) Numerical aperture (3) Acceptance angle                                                                                                                        | <b>04</b>    |
|            | <b>(c)</b> With figure explain The Modified Chemical Vapour Deposition (MCVD) method of fiber fabrication in detail. What are its advantages and disadvantages?                                                                                                                                                  | <b>07</b>    |
|            | <b>OR</b>                                                                                                                                                                                                                                                                                                        |              |
| <b>Q.3</b> | <b>(a)</b> A step index fiber in air has a NA of 0.16, a core refractive index of 1.45 and a core diameter of 60 $\mu\text{m}$ . Determine the normalized Frequency for the fiber when light at a wavelength of 0.82 $\mu\text{m}$ is transmitted. Estimate the number of guided modes propagating in the fiber. | <b>03</b>    |
|            | <b>(b)</b> Draw and Describe the structure of Planar LED.                                                                                                                                                                                                                                                        | <b>04</b>    |
|            | <b>(c)</b> With figure explain The Plasma Chemical Vapour Deposition (PCVD) method for the production of an optical fiber. What are its advantages?                                                                                                                                                              | <b>07</b>    |
| <b>Q.4</b> | <b>(a)</b> Illustrate the forward and backward pumping capability associated with fiber Raman amplifier.                                                                                                                                                                                                         | <b>03</b>    |
|            | <b>(b)</b> For a 30 km long fiber the attenuation is 0.8 dB/km at a wave length of 1300nm. If a 200 $\mu\text{W}$ power is launched into the fiber, find the output power.                                                                                                                                       | <b>04</b>    |

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|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
|     | (c) | Define scattering losses and explain linear scattering losses in detail.                                                                                                                                                                                                                                                                                         | 07 |
|     |     | <b>OR</b>                                                                                                                                                                                                                                                                                                                                                        |    |
| Q.4 | (a) | Explain stimulated emission in LASER.                                                                                                                                                                                                                                                                                                                            | 03 |
|     | (b) | Explain the spontaneous emission in LASER with necessary diagram.                                                                                                                                                                                                                                                                                                | 04 |
|     | (c) | Describe The principle, characteristics and operation of Avalanche photo Diode in detail.                                                                                                                                                                                                                                                                        | 07 |
| Q.5 | (a) | Draw the schematic structure of the semiconductor optical amplifier and explain its principle.                                                                                                                                                                                                                                                                   | 03 |
|     | (b) | Give the comparison between Edge Emitter and surface emitter LEDs                                                                                                                                                                                                                                                                                                | 04 |
|     | (c) | What are the bending losses in fiber optic communication?<br>Describe: (1) Micro bending losses (2) Macro bending losses.                                                                                                                                                                                                                                        | 07 |
|     |     | <b>OR</b>                                                                                                                                                                                                                                                                                                                                                        |    |
| Q.5 | (a) | Explain the concept of Wavelength Division Multiplexed Networks (WDM) briefly.                                                                                                                                                                                                                                                                                   | 03 |
|     | (b) | The radiative and nonradiative recombination lifetimes of the minority carriers in the active region of a double-heterojunction LED are 60ns and 100ns respectively. Determine the total carrier recombination lifetime and the power internally generated within the device when the peak emission wavelength is 0.87 $\mu\text{m}$ at a drive current of 40mA. | 04 |
|     | (c) | With figures describe the structure and function of a surface emitting LED. What are the advantages and disadvantages of surface emitting LED?                                                                                                                                                                                                                   | 07 |

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